Earthquakes & Volcanoes

CASE STUDY | Japan 2011 Tohoku Earthquake (MEDC)

CASE STUDY | Nevada del Ruiz Volcano, Colombia 1985 (LEDC)

SYLLABUS OBJECTIVES | What you need to know

- Structure of the Earth and tectonic plates
- Three types of plate boundary and processes at each
- Causes of earthquakes and volcanic eruptions
- Features of volcanoes (composite and shield) and earthquakes
- Effects on people and the environment (social, economic, environmental)
- Volcanoes as hazards AND opportunities for people
- Strategies to reduce the impacts of earthquakes and volcanoes
- Case study: Japan 2011 Tohoku Earthquake (MEDC)
- Case study: Nevada del Ruiz Volcano, Colombia 1985 (LEDC)

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1. Structure of the Earth

The Earth is made up of four main layers. Understanding these layers is the foundation of plate tectonics.

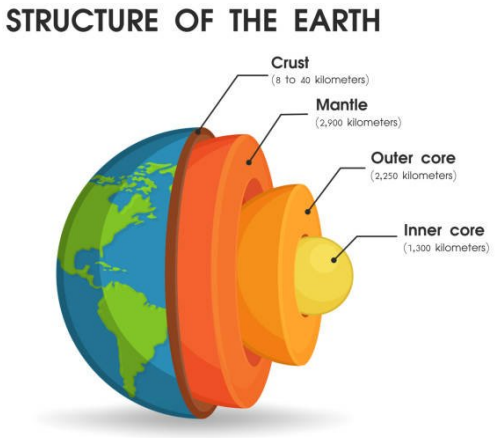

Fig 1.1 — Cross-section showing the four layers of the Earth

Layer	Depth	State	Key Facts
Inner Core	0–1,300 km	Solid	Iron and nickel; ~5,000°C; solid due to immense pressure
Outer Core	1,300–3,550 km	Liquid	Liquid iron and nickel; generates Earth's magnetic field
Mantle	3,550–6,370 km	Semi-molten	Largest layer; convection currents drive plate movement
Crust	6–40 km thick	Solid	Oceanic: thin, dense, basalt. Continental: thicker, less dense, granite

THE LITHOSPHERE

The lithosphere = crust + uppermost solid mantle. It is broken into tectonic plates.
Convection currents in the mantle cause the plates to move 2–10 cm per year.
Where plates meet = a plate boundary (also called a plate margin).

2. Global Pattern of Tectonic Plates

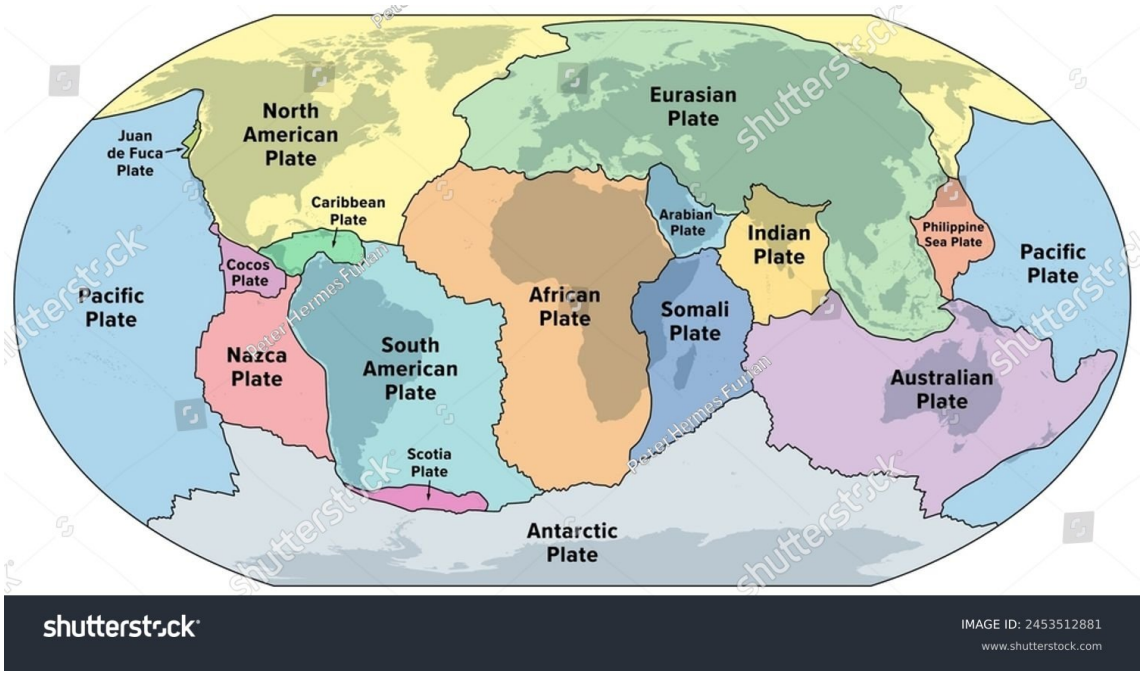

Fig 2.1 — World map of major and minor tectonic plates

Major Tectonic Plates — Names and Locations

Major Plate	Type	Minor Plate	Type
Pacific	Oceanic	Caribbean	Oceanic
North American	Continental	Cocos	Oceanic
South American	Continental	Nazca	Oceanic
Eurasian	Continental	Arabian	Continental
African	Continental	Philippine	Oceanic
Indian	Continental	Scotia	Oceanic
Australian	Continental	Juan de Fuca	Oceanic
Antarctic	Cont./Oceanic mix	Somali	Continental

3. Plate Boundary Types

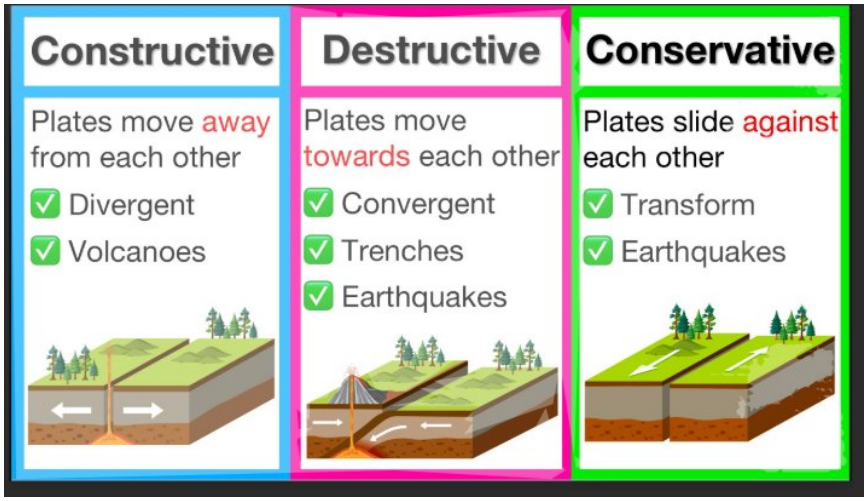

Fig 3.1 — The three types of plate boundary

Boundary	Movement	Process	Landforms / Hazards	Example
Constructive (Divergent)	Plates move apart	Magma rises and fills the gap; new crust forms	Mid-ocean ridges, rift valleys, shield volcanoes, mild earthquakes	Mid-Atlantic Ridge; East African Rift
Destructive (Convergent)	Plates collide	Denser oceanic plate subducts beneath continental; melts to magma	Oceanic trenches, composite volcanoes, fold mountains, powerful earthquakes	Nazca–S. America: Andes; Pacific–N. America: Cascades
Conservative (Transform)	Plates slide past each other	Plates lock then jolt; no crust created or destroyed	Fault lines, violent earthquakes, **NO volcanoes**	San Andreas Fault; N. Anatolian Fault

4. Types of Volcano

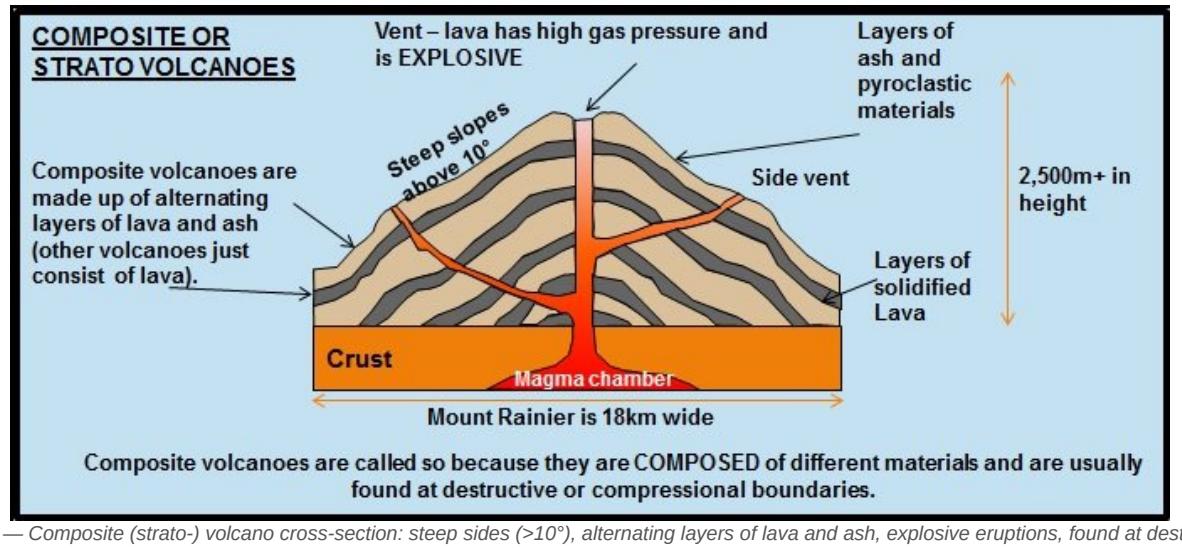

Fig 4.1 — Composite (strato-) volcano cross-section: steep sides (>10°), alternating layers of lava and ash, explosive eruptions, found at destructive boundaries

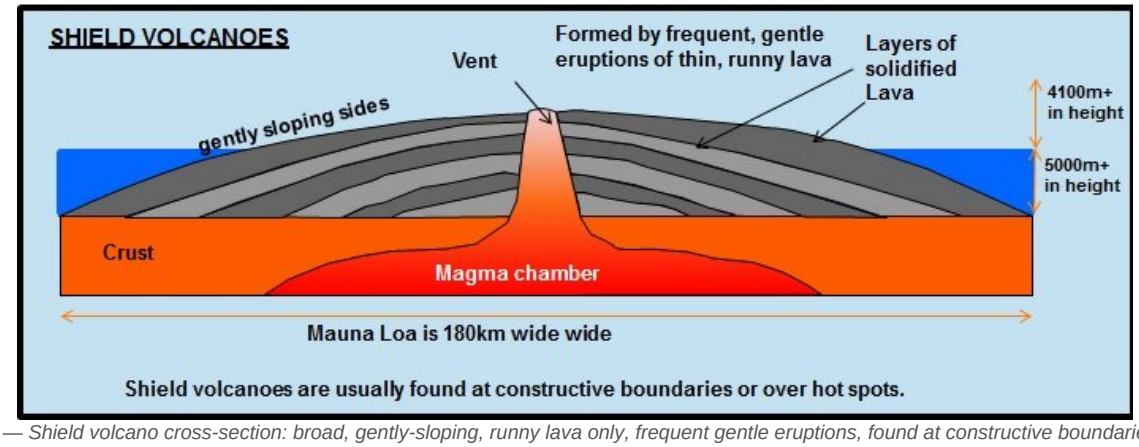

Fig 4.2 — Shield volcano cross-section: broad, gently-sloping, runny lava only, frequent gentle eruptions, found at constructive boundaries or hot spots

Feature	Composite (Strato-) Volcano	Shield Volcano
Shape	Steep cone, sides >10°	Broad dome, gentle slopes
Height	Up to 2,500 m+	Up to 4,100 m+ but very wide (Mauna Loa: 180 km)
Lava	Thick, viscous, high gas pressure	Thin and runny (low viscosity)
Eruption	Explosive, violent, infrequent	Gentle, frequent, effusive
Made of	Alternating layers of lava **AND** ash	Layers of solidified lava only
Boundary	Destructive / convergent	Constructive / hot spot
Example	Nevada del Ruiz; Mt Fuji; Mt St Helens	Mauna Loa, Hawaii; Iceland

4b. Volcano Structure — Key Features

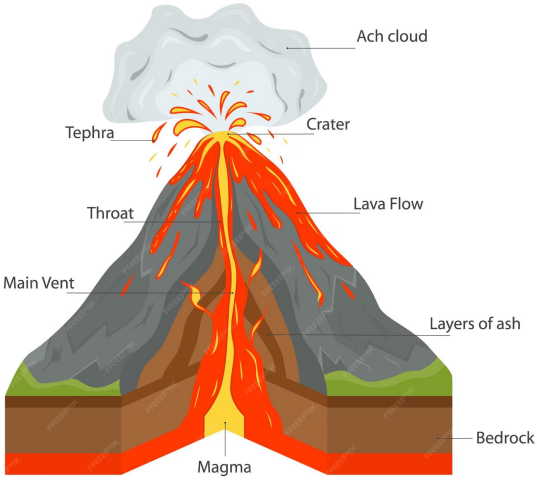

Fig 4.3 — Labelled volcano cross-section. Note: 'Throat' = VENT (Cambridge term). 'Ach cloud' = ASH cloud.

Feature	Definition	Feature	Definition
Crater	Bowl-shaped hollow at the volcano summit	**Ash Cloud**	Cloud of fine particles and gas blasted from the crater
Vent	The pipe/channel through which magma reaches the surface	**Tephra**	All solid material ejected: ash, pumice, volcanic bombs
Magma Chamber	Underground reservoir of molten rock below the volcano	**Pyroclastic flow**	Superheated mix of gas, ash and rock (700°C) rushing downslope
Lava Flow	Magma that erupts and flows down the volcano flanks	**Lahar**	Volcanic mudflow: ash mixes with meltwater or rain

5. Features and Causes of Earthquakes

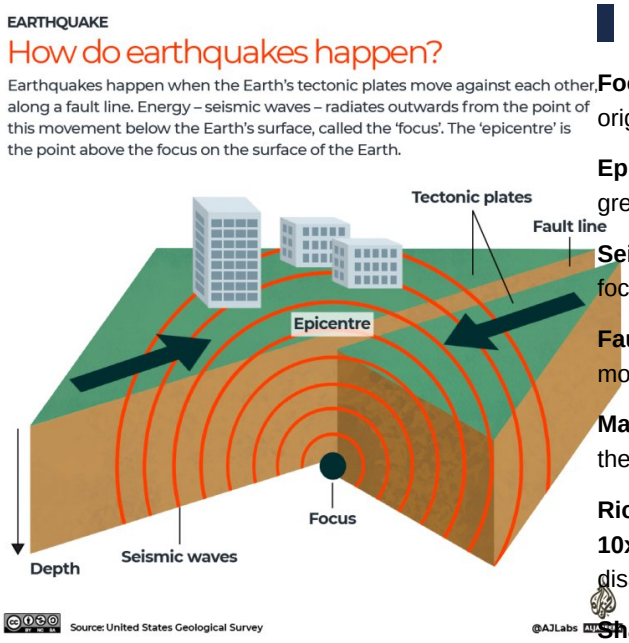

Fig 5.1 — Focus, epicentre and seismic waves

Key Earthquake Vocabulary

Focus (Hypocentre) — Point underground where earthquake originates; energy released here.

Epicentre — Point on the surface directly above the focus; greatest damage here.

Seismic Waves — Shock waves radiating outward from the focus. P-waves (primary, faster) and S-waves (secondary).

Fault Line — A fracture in the Earth's crust where plates have moved.

Magnitude — The size/energy of an earthquake, measured on the Richter Scale.

Richter Scale — Logarithmic scale 1–10+. Each whole number = **10x** more ground movement. Below 4 = minor; above 7 = major disaster.

Shallow focus (0–70 km) = most destructive. **Deep focus** (>300 km) = energy lost before reaching surface.

Causes of Earthquakes

• Tectonic plates move due to **convection currents** in the mantle.

• At **conservative boundaries**: plates lock and then suddenly jolt forward, releasing huge amounts of energy as seismic waves — the most common cause of major earthquakes.

• At **destructive boundaries**: friction as one plate subducts beneath the other triggers earthquakes.

• At **constructive boundaries**: plates pull apart, causing smaller, shallower earthquakes.

• A **tsunami** can be triggered when a large submarine earthquake displaces the sea floor.

6. Effects on People and the Environment

Type	Social	Economic	Environmental
Primary (immediate)	Deaths and injuries; buildings collapse; people trapped; homelessness	Businesses and infrastructure destroyed; immediate rescue costs	Ground cracking; liquefaction; landslides; tsunamis triggered
Secondary (longer-term)	Disease from polluted water; mental health trauma; displaced communities	Long-term rebuild costs; loss of tourism; unemployment	Fires; soil and water contamination; habitat destruction; nuclear risk

7. Volcanoes — Hazards AND Opportunities for People

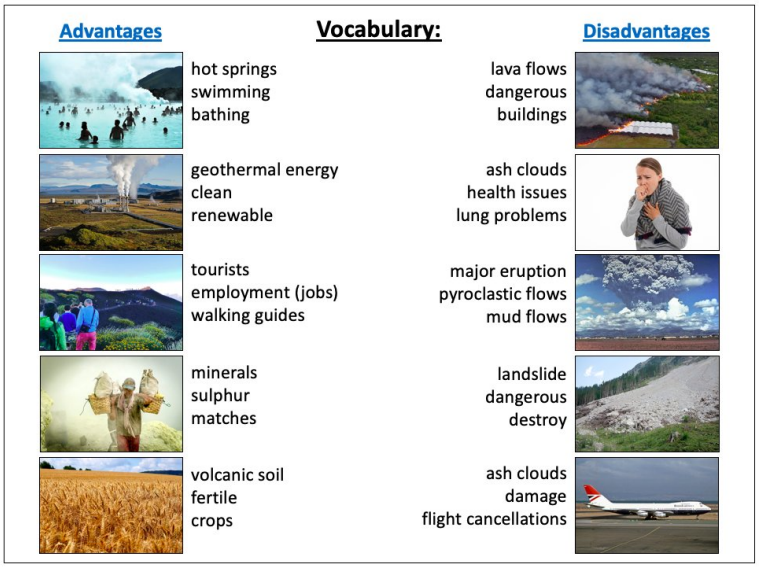

Fig 7.1 — Advantages (left) and disadvantages (right) of living near a volcano

HAZARDS (disadvantages)	OPPORTUNITIES (advantages)
• **Lava flows** destroy homes, farmland and infrastructure • **Pyroclastic flows**: superheated gas/ash (700°C+), kills instantly • **Ash clouds**: disrupt flights, cause lung disease, collapse roofs • **Lahars** (volcanic mudflows) bury entire towns • **Landslides** triggered by eruption/earthquakes • **Tsunamis** triggered by underwater volcanic activity	• **Fertile soil**: volcanic ash enriches soil — excellent for farming • **Geothermal energy**: clean, renewable heat energy (e.g. Iceland) • **Tourism**: scenic landscapes attract millions of visitors and jobs • **Minerals**: sulphur, gold, copper, diamonds found near volcanoes • **Hot springs**: bathing, spas and health tourism (Blue Lagoon, Iceland) • **Employment**: guides, scientists, engineers in volcanic regions

8. Reducing the Impacts of Earthquakes and Volcanoes

Strategy	Description	Example
Prediction and Monitoring	Seismometers detect tremors; tiltmeters measure ground swelling; satellite GPS monitors plate movement; gas emission sensors near volcanoes	Japan Meteorological Agency 24/7 monitoring; Montserrat Volcano Observatory
Planning and Preparedness	Evacuation routes and regular drills; stockpiling emergency supplies; early warning systems; trained rescue teams on standby	Japan's J-Alert system gave warnings before 2011 shaking; Montserrat exclusion zones
Land Use Zoning	Hazard maps identify high-risk areas; building restricted near volcanoes or active fault lines; low-risk zones for dense housing	Montserrat: south exclusion zone, north 'safe zone'; San Francisco strict zoning along San Andreas Fault
Building Design	Earthquake-resistant construction: cross-bracing, rubber shock absorbers, reinforced concrete, counterweight pendulums on top floors	Japan: skyscrapers with pendulum counterweights; traditional pagodas designed to flex not crack
Education	School earthquake drills; public 'drop, cover, hold on' campaigns; community first aid training; tsunami evacuation signage	Japan: mandatory school drills since 1960s; US National ShakeOut drills

9. Case Study 1 — Japan 2011 Tohoku Earthquake and Tsunami (MEDC)

FAST FACTS — JAPAN 2011

Date: 11 March 2011 | Magnitude: 9.0 Mw (4th strongest earthquake ever recorded)
Location: Pacific Ocean, 70 km off the east coast of Tohoku, Japan
Boundary type: Pacific Plate subducting beneath the North American Plate (DESTRUCTIVE)
Focus depth: 32 km — relatively shallow, making shaking more intense at the surface
Tsunami: waves up to 40 m high; travelled up to 10 km inland in some areas

Fig 9.1 — Devastation in Tohoku: entire communities swept away by the tsunami

Causes

- The Pacific Plate moves north-westward at approximately 8 cm per year, subducting beneath the North American Plate.
- The two plates had been locked together for around 1,000 years, building up enormous stress.
- A 500 km section of the fault suddenly ruptured, releasing energy equivalent to 600 million Hiroshima bombs.
- The seafloor lurched upward by up to 7 m, displacing a massive column of ocean water — triggering the tsunami.
- The first tsunami waves struck the Tohoku coast within 10–30 minutes of the earthquake.

Social Impacts	Economic Impacts	Environmental Impacts
• Approx. 15,900 killed; 2,500+ still missing • 450,000+ displaced or made homeless • Fukushima Daiichi nuclear disaster — radiation evacuations • Hospitals overwhelmed; many destroyed • Water and electricity cut off for weeks	• Estimated cost: US$235 billion — costliest disaster in history • 4.4 million households without electricity • Toyota and Sony factories shut for months • Ports and fishing industry destroyed • Japan's GDP fell by 0.7%	• Tsunami scoured 561 km² of coastline • 24 million tonnes of debris left behind • Soil liquefaction widespread inland • Coastline dropped by up to 1.2 m permanently • Nuclear contamination of soil and Pacific Ocean

Responses

Immediate (Short-term)	Long-term
• J-Alert gave 1 minute warning before main shaking • 100,000 military deployed within 24 hours • International aid from 116 countries received • 20 km nuclear exclusion zone around Fukushima • Emergency shelters set up in schools and halls	• US$235 billion reconstruction programme launched • Massive sea walls built (up to 15 m high) along coastline • Fukushima clean-up — expected to take 40 years • 48 nuclear plants temporarily shut for safety review • New laws requiring tsunami-resistant buildings in coastal zones

WHY WAS THE IMPACT REDUCED? (MEDC factors)

Strict earthquake-resistant building codes (upgraded 1981, 2000) meant fewer building collapses.
J-Alert early warning system gave people precious seconds to take cover.
Regular mandatory earthquake drills — the population knew exactly what to do.
Well-funded, well-equipped emergency services and military able to respond at scale.
BUT: even Japan could not fully prepare for a 9.0 earthquake combined with 40 m tsunami waves.

10. Case Study 2 — Nevada del Ruiz, Colombia 1985 (Volcano / LEDC)

FAST FACTS — NEVADA DEL RUIZ 1985

Date: 13 November 1985 | Volcano type: Composite (strato-) volcano
Location: Colombian Andes, 5,321 m high — permanently snow- and ice-capped at summit
Boundary: Nazca Plate subducting beneath South American Plate (DESTRUCTIVE boundary)
Eruption size: Moderate (VEI 3) — but triggered catastrophic LAHARS (volcanic mudflows)
Death toll: approximately 23,000 — 4th deadliest volcanic disaster in recorded history

Fig 10.1 — Volcanic eruption plume rising above a city, similar to the Nevada del Ruiz eruption that sent lahars racing down valleys at 60 km/h

Causes

• The **Nazca Plate** subducts beneath the **South American Plate** at a destructive boundary, generating magma that feeds composite volcanoes along the Andes.

• The eruption was **moderate in size** — it was the **lahars** (not lava) that caused mass death.

• Heat from the eruption **rapidly melted the summit glaciers** and snowfields, releasing enormous volumes of water.

• Water mixed with volcanic ash and debris, forming lahars that surged down river valleys at up to **60 km/h**.

• The lahar travelled **100 km** down the Rio Lagunillas valley and buried the town of **Armero** under 5 m of mud.

• Armero had a population of 29,000. The lahar arrived at **11:35 PM** — most residents were asleep.

Social Impacts	Economic Impacts	Environmental Impacts
• Approx. 23,000 killed (out of 29,000 in Armero) • 5,000+ injured • 8,000+ made homeless • Armero entirely destroyed • Widespread trauma; survivors had lost everything	• Cost estimated at ~US$1 billion (7.7% of Colombia's GDP) • Agricultural land buried under mud • Roads and bridges destroyed • Colombia too poor for rapid independent recovery • Aid received from 43 countries	• Armero valley buried under deep volcanic mud • River channels blocked and redirected • 200 km² of agricultural land destroyed • Water supplies contaminated • Long-term soil and ecosystem damage

Responses and Why the Death Toll Was So High

Responses	Why so many died — LEDC factors
• Colombian Red Cross and military deployed • International rescue teams from many countries • Survivors airlifted from rooftops by helicopter • Emergency medical centres and food distribution • Armero never rebuilt — became a memorial site	Warning ignored: a hazard map was published 2 months earlier but not distributed Evacuation cancelled: officials thought the danger had passed — residents returned home No early warning: no sirens reached Armero before the lahar LEDC poverty: limited funds for monitoring equipment and emergency services Night-time: eruption at 11:35 PM — most people were asleep in their homes

11. Key Vocabulary — 20 Terms You Must Know

Term	Definition
Tectonic plate	A large slab of the Earth's lithosphere that moves on the semi-molten mantle.
Plate boundary	Where two tectonic plates meet; the site of most earthquakes and volcanic eruptions.
Constructive boundary	Plates move apart; magma rises and fills the gap; new oceanic crust is formed.
Destructive boundary	Plates collide; denser oceanic plate subducts beneath continental plate; melts to magma.
Conservative boundary	Plates slide past each other; no crust created or destroyed; major earthquakes occur.
Subduction	When a denser oceanic plate sinks beneath a less dense continental plate into the mantle.
Magma / Lava	Molten rock below the surface (magma); called lava once it erupts at the surface.
Composite volcano	Steep-sided cone of alternating lava and ash; explosive eruptions; destructive boundary.
Shield volcano	Broad, gently-sloping; runny lava only; gentle eruptions; constructive/hot spot.
Focus (hypocentre)	Point underground where an earthquake originates and energy is first released.
Epicentre	Point on the Earth's surface directly above the focus; greatest damage usually here.
Seismic waves	Shock waves that radiate outward in all directions from the earthquake focus.
Richter scale	Logarithmic scale measuring earthquake magnitude. Each unit = 10x more ground movement.
Lahar	Volcanic mudflow; formed when eruption melts ice/snow or mixes with rain water.
Pyroclastic flow	Superheated mix of gas, ash and rock (up to 700°C) rushing rapidly down a volcano slope.
Liquefaction	Water-saturated soil behaves like liquid during shaking; buildings sink or topple.
Tsunami	Series of giant ocean waves triggered by seafloor earthquake, landslide or eruption.
Convection current	Circular heat movement in the mantle — the driving force behind plate movement.
Geothermal energy	Renewable energy extracted from underground heat near volcanic areas (e.g. Iceland).
Hazard map	A map showing areas at risk from a natural hazard; used for planning and evacuation.

12. Syllabus Checklist — Cambridge 0460 Theme 2.1

- ■ Describe the structure of the Earth: inner core, outer core, mantle, crust, lithosphere
- ■ Explain how convection currents drive plate movement
- ■ Name and locate the major tectonic plates on a world map
- ■ Describe processes and landforms at constructive, destructive and conservative plate boundaries
- ■ Explain the causes of earthquakes at different boundary types
- ■ Define: focus, epicentre, seismic waves, magnitude, Richter scale
- ■ Describe and compare features of composite and shield volcanoes
- ■ Explain how composite and shield volcanoes are formed
- ■ Describe social, economic and environmental effects of tectonic hazards
- ■ Explain volcanic hazards: lava flows, pyroclastic flows, lahars, ash clouds, tsunamis
- ■ Explain volcanic opportunities: fertile soil, geothermal energy, tourism, minerals, hot springs
- ■ Describe 5 strategies to reduce impacts: prediction, planning, zoning, building design, education
- ■ CASE STUDY: Japan 2011 — boundary type, causes, social/economic/environmental effects, responses

- CASE STUDY: Nevada del Ruiz 1985 — causes (lahar mechanism), effects, why death toll so high
- Explain why MEDC responses differ from LEDC responses with reference to both case studies
- Use specific data and statistics to support exam answers

13. Exam Alert — Common Mistakes and Mark-Winning Tips

COMMON MISTAKES TO AVOID

Do NOT write 'plates moving' — name the boundary type AND the plates involved.
Do NOT confuse focus (underground) and epicentre (on the surface).
Do NOT say volcanoes form at ALL boundaries — conservative boundaries have NO volcanoes.
Do NOT mix up composite and shield volcanoes — learn slope, lava type and boundary type.
Do NOT forget that LAHARS killed most people at Nevada del Ruiz — not lava flows.
Do NOT just say 'MEDC = less deaths' — explain WHY (building codes, warnings, money, training).
Do NOT forget that volcanoes offer OPPORTUNITIES — the examiner will always test this.

MARK-WINNING TIPS

Always use specific data: '9.0 on the Richter Scale'; '23,000 killed at Armero'; 'US$235 billion'.
Name both case studies explicitly in every extended answer — never just 'a country' or 'an LEDC'.
For EXPLAIN questions: use connectives — 'because', 'therefore', 'this means that', 'as a result'.
For 4-mark questions: write 2 well-developed points, not 4 thin bullet points.
For 7-mark questions: use PEEL — Point, Evidence, Explain, Link back to the question.
Spell key terms correctly: LAHAR, PYROCLASTIC, EPICENTRE, RICHTER — examiners notice.
If asked to 'describe and explain effects': give at least one social AND one economic point.

14. Exam Practice Questions — Cambridge 0460 Style

All questions follow the Cambridge 0460 Paper 1 command word style. Section B questions on tectonic hazards use the 1 + 2 + 3 + 4 + 7 structure.

Question 1

[1 marks]

(a) What is the term for the point on the Earth's surface directly above the focus of an earthquake? [1]

MARK SCHEME

1 mark: Epicentre (accept: epicenter) ✓

Question 2

[2 marks]

(b) State two ways in which living near a volcano can be an advantage for people. [2]

MARK SCHEME

1 mark each for any two of the following (up to 2 marks):

Fertile volcanic soil for farming ✓ | Geothermal energy (clean/renewable) ✓

Tourism attraction, jobs as guides ✓ | Minerals: sulphur, gold, copper ✓

Hot springs / bathing / health tourism ✓

Question 3

[3 marks]

(c) Explain the differences between a constructive and a destructive plate boundary. [3]

MARK SCHEME

Award 1 mark per clear, correct, developed point (max 3):

Constructive: plates move apart ✓; magma rises to fill the gap / new crust formed ✓; shield volcanoes / mild EQs ✓

Destructive: plates collide ✓; denser oceanic plate subducts / melts ✓; composite volcanoes / powerful EQs / trenches ✓

Question 4

[4 marks]

(d) For a named tectonic hazard you have studied, describe and explain its effects on people. [4]

Named hazard: _____

MARK SCHEME

Credit any well-developed social or economic effects with accurate detail (max 4 marks):

Japan 2011 examples — Social: 15,900 killed ✓; 450,000 displaced ✓; Fukushima nuclear evacuations ✓

Economic: US$235bn cost ✓; Toyota/Sony factories closed ✓; GDP fell 0.7% ✓

Nevada del Ruiz — Social: 23,000 killed at Armero ✓; 8,000 homeless ✓

Economic: US$1bn cost / 7.7% of Colombia's GDP ✓; agricultural land buried ✓

Must name the hazard. Max 2 marks for social; max 2 for economic — must cover both for full marks.

Question 5 — Extended Writing [7 marks]

[7 marks]

(e) For a named volcanic eruption that you have studied, describe and explain the effects of the eruption on people and the environment. [7]

Named volcanic eruption: _____

MARK SCHEME

Level 3 [6–7]: Detailed description AND explanation of effects on BOTH people and environment; accurate named data; clear cause-and-effect reasoning; well-structured response.
e.g. Nevada del Ruiz: lahar mechanism; 23,000 killed; night-time; warning ignored; 200 km2 farmland buried.

Level 2 [4–5]: Effects described with some explanation; covers mainly people OR environment; some data used.

Level 1 [1–3]: Basic description only; general statements; limited or no named detail.

Accept Japan 2011 tsunami/earthquake — credit any named volcanic OR earthquake case study fully.

PEEL FRAMEWORK — use this structure for ALL 7-mark answers

P	**POINT**	Open with a clear geographical statement answering the question directly.
E	**EVIDENCE**	Support with specific named data — deaths, costs, dates, place names.
E	**EXPLAIN**	Explain the geographical reason WHY — use 'because', 'therefore', 'this means'.
L	**LINK**	Link back to the question wording. Develop the point or add a second example.

15. Summary Card — Cut Out and Revise

PLATE BOUNDARIES	VOLCANO TYPES	EARTHQUAKE FEATURES

Constructive: plates apart → magma fills gap → new crust → shield volcanoes Destructive: plates collide → subduction → composite volcanoes + trenches Conservative: plates slide past → earthquakes ONLY, no volcanoes	COMPOSITE: steep sides, layers of lava + ash, explosive, destructive boundary Examples: Nevada del Ruiz; Mt Fuji SHIELD: broad, gentle, runny lava only, frequent/gentle, constructive/hot spot Examples: Mauna Loa; Iceland	Focus = underground origin Epicentre = surface point above focus Seismic waves = energy outward Richter scale (logarithmic) Shallow focus = MORE destructive Conservative = worst earthquakes

JAPAN 2011 (MEDC)	NEVADA DEL RUIZ 1985 (LEDC)	REDUCING IMPACTS
Boundary: Pacific → N. American (destructive) Magnitude 9.0; tsunami up to 40 m high ~15,900 killed; US$235 billion damage MEDC: building codes, J-Alert, drills Long-term: sea walls, Fukushima clean-up	Boundary: Nazca → S. American (destructive) Lahar buried Armero at 11:35 PM ~23,000 killed; US$1 billion cost LEDC: warning ignored, no sirens, poverty Deaths were largely preventable	1. Prediction and monitoring 2. Planning and evacuation drills 3. Land use zoning / hazard maps 4. Earthquake-resistant building design 5. Public education and awareness

EXAM REMINDER

Cambridge 0460 Paper 1 Section B: 1 + 2 + 3 + 4 + 7 mark structure for this topic.
Always name your case study: 'Nevada del Ruiz 1985' / 'Japan 2011' — never just 'a volcano'.
The 7-mark answer needs: description + explanation + named data + conclusion.
Revise: plate boundary diagrams | volcano features | 5 impact-reduction strategies.